**Code No: 111AH** **R13**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B.Tech I Year Examinations, July - 2021**
**ENGINEERING DRAWING**
**(Common to CSE, MIE, PTM)**

**Time: 3 hours** **Max Marks: 75**

**Answer any three questions**
**All questions carry equal marks**
**- - -**

1.a) The actual length of 300 m is represented by a line of 10 cm on a drawing. Draw a vernier scale to read up to 500 m. Mark on it a length of 367 m.
b) Draw a regular heptagon having 40 mm long sides, using general method. [15+10]

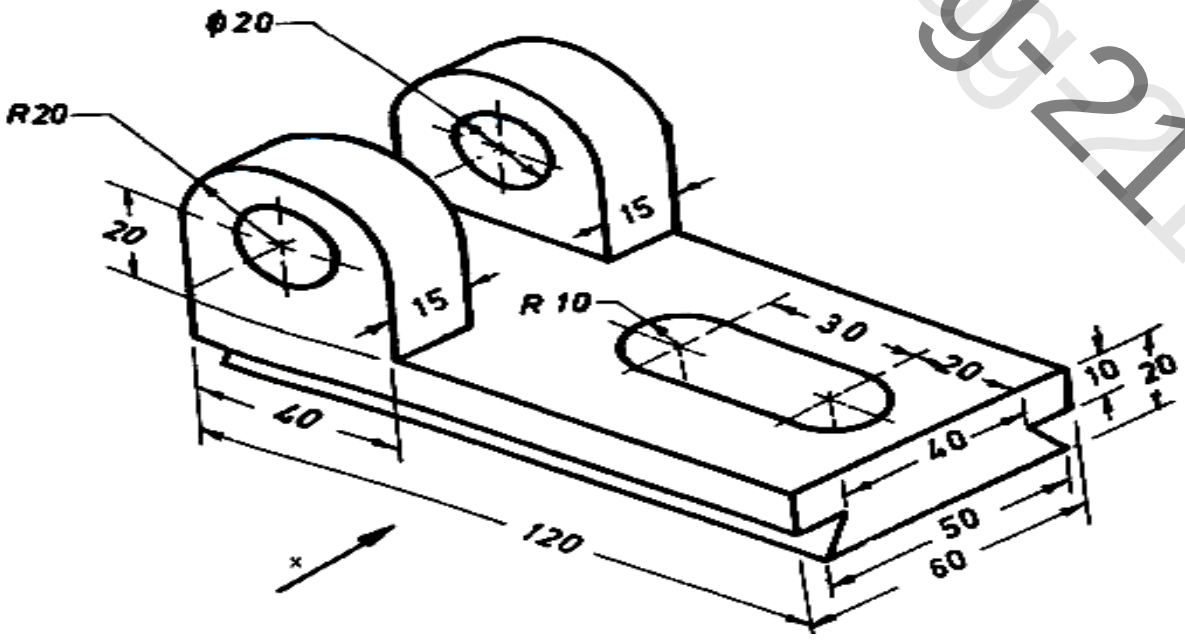
2. Draw the locus of a point P moving so that the ratio of the distance from a fixed point F to its distance from a fixed straight line is 0.75. Point P is at a distance of 70 mm from the straight line. Also, draw the tangent and normal to the curve. [25]

3. A circular plate of 60 mm diameter has a hexagonal hole of 20 mm side, centrally punched. Draw the projections of the plate, resting on HP on a point, with its surface inclined at $30^0$ to HP. Any two parallel sides of the hexagonal hole are perpendicular to VP. Draw the projections of the plate. [25]

4. A 90 mm long line PQ is inclined at $30^0$ to the HP and $45^0$ to the VP. The end P is 15 mm above the HP and 25 mm in front of VP. Draw its projections, locate the traces in the following two cases.
a) When the end Q is in the first quadrant
b) When the end Q is in the third quadrant. [15+10]

5. A cone of base diameter 50 mm and axis length 75 mm, resting on HP on its base is cut by a plane inclined at $45^0$ to HP and perpendicular to VP and is bisecting the axis. Draw the front view and development of lateral surface of bottom part. [25]

6. Draw the front view, top view and side view of the object shown in the figure. All dimensions are in mm. [25]

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